

Paper 1**AI-Assisted Portfolio Triage for Canada's Major Projects: Evidence from the Major Projects Inventory, 2018–2024****Running Title:** AI Portfolio Triage – Evidence from MPI**Author:** Joshua Samuel**Affiliation:** Northeast Renewables LP, Toronto, Ontario, Canada**Contact Information:** Email: jsamuel@northeastmidstream.com**Keywords:** Major project development; Canadian infrastructure policy; Regulatory bottlenecks; Project success rates; Capital deployment; Investment incentives**Word Count:** ~2,300**Date:** August 2025**Funding Statement:** This research received no external funding.**Conflict of Interest:** The author declares no conflict of interest.

Abstract

Major infrastructure development in Canada plays a central role in achieving national economic and climate goals. While past research has explored barriers to major project delivery, most analyses remain fragmented across sectors or regions. The lack of cross-cutting, longitudinal data has limited understanding of structural bottlenecks and policy impacts. This study uses descriptive statistics, cohort tracking, and status transition analysis on 579 projects from the Major Projects Inventory (2018–2024) to evaluate project advancement and outcomes. Results show: (i) only 19 percent of projects in development reach construction; (ii) early-stage attrition is steep, with 66 percent of planning-stage projects never progressing; (iii) the mining sector and projects in British Columbia face particularly low success rates; (iv) post-approval delays are common, with over \$14 billion tied up in approved projects stuck for 3+ years; and (v) recent policy tools such as investment tax credits and public investment funds appear to stimulate new project announcements and increase project size. Continued tracking of newer cohorts is needed to determine whether this momentum converts into higher progression rates or if more focused measures are still required to clear persistent bottlenecks and fully realize Canada's infrastructure potential.

Introduction

Canada's Major Projects Inventory or MPI maintained by Natural Resources Canada offers a unique view of the major project pipeline in Canada. Projects currently in the MPI represent roughly 20 per cent of Canada's gross fixed capital formation over the next decade and are central to meeting national targets for critical minerals, clean fuels, and net-zero electricity. Yet only 109 of the 579 (19 percent) projects in our 2018–2024 dataset are either under construction or have reached a completed state by 2024. Understanding where, when, and why projects either proceed to construction or fall away can inform more effective policy design, de-risk private capital, and shorten the path from conceptual design to in-service asset.

The goal of this paper is to use the MPI dataset to answer three questions. First, what proportion of major projects advance to construction, and how quickly? Second, how do outcomes differ by sector, province, project size and ownership? Third, what has been the effect of major federal policy initiatives and financial incentives on project development, although we may not be able to answer this question with any certainty, as such policies often take considerable time to flow through and their effects can be difficult to isolate from other market and project-specific factors?

Methods

The full MPI dataset comprises 1,250 major Canadian capital projects tracked from 2017 to 2024. It includes projects across multiple sectors (Energy, Mining, Forestry) and all provinces and territories, with project costs ranging from \$20 million to over \$40 billion. Each project record contains information on project status at multiple time points, geographic location, sector classification, estimated costs, and key project characteristics.

The MPI categorizes projects by development stage as follows: “Announced & Planning” refers to projects that have been publicly disclosed but have not yet submitted regulatory applications; “In Review” includes projects with submitted applications that are currently under assessment; and “Approved” applies to projects that have received all major regulatory approvals required to begin construction but have not yet started building. The final stage, “Under Construction,” includes projects where physical construction is underway.

For the purposes of this study, two systematic exclusions were applied to create the analytical cohort, reducing the dataset from 1,250 to 579 projects (54 percent reduction). First, all projects with 2017 as their start year were excluded from the primary analysis as 2017 represents a baseline year with incomplete observational windows and these projects may have been in progress before systematic tracking began. Second, projects with “Under Construction” as their initial status were excluded as they exhibited fundamentally different risk profiles compared to the others, with 94 percent maintaining successful status throughout the observation period. By focusing on projects beginning in Announced & Planning, In Review, or Approved stages, the analysis better captures factors that determine whether projects advance to construction. The resulting analytical cohort of 579 projects provides a consistent, representative basis for examining advancement patterns, with sufficient temporal scope to observe complete development cycles. This cohort maintains representativeness across provinces and sectors while providing sufficient observation time to track project progression through regulatory and development phases.

For this study, cohorts are grouped by entry year. For each cohort we calculate success rate and median years to construction. Sub-group analyses segment outcomes by sector (Energy, Mining, Forest), province, cost quintile and ownership type. To probe non-regulatory bottlenecks, we flag

projects that remained in “Approved” status for three or more years. Policy period comparisons divide the timeline into four eras aligning with the introduction of major federal policy and incentive initiatives.

Success is defined as a project that had started as Announced & Planning, In Review, or Approved and then transitioned to Under Construction. This binary definition does not account for partial completions, phased operations, or project quality. Results may also be biased by incomplete observation windows for projects and potential reporting uncertainties.

Summary tables underpin province, sector, and cost and ownership analyses. Simple logistic regressions estimate the odds of success by sector and province, controlling for cost and entry year. Kaplan-Meier survival curves track time-to-approval, and Kruskal-Wallis tests probe differences across policy periods. All statistics are descriptive. No external data or re-estimations were introduced.

Results & Discussion

Cohort progression

Figure 1 summary statistics by entry year cohort show a clear temporal pattern: earlier cohorts (2018–2020) exhibit higher success rates (26–31 percent) and longer average durations, while more recent cohorts (2021–2023) show lower success rates and shorter timelines, reflecting limited time for progression. The 2024 cohort has no completions to date due to this lag effect. The lower success rates seen in the newer cohorts appear to be largely due to timing effects. Additional research is needed to confirm whether any real performance gaps exist between the earlier and more recent cohorts.

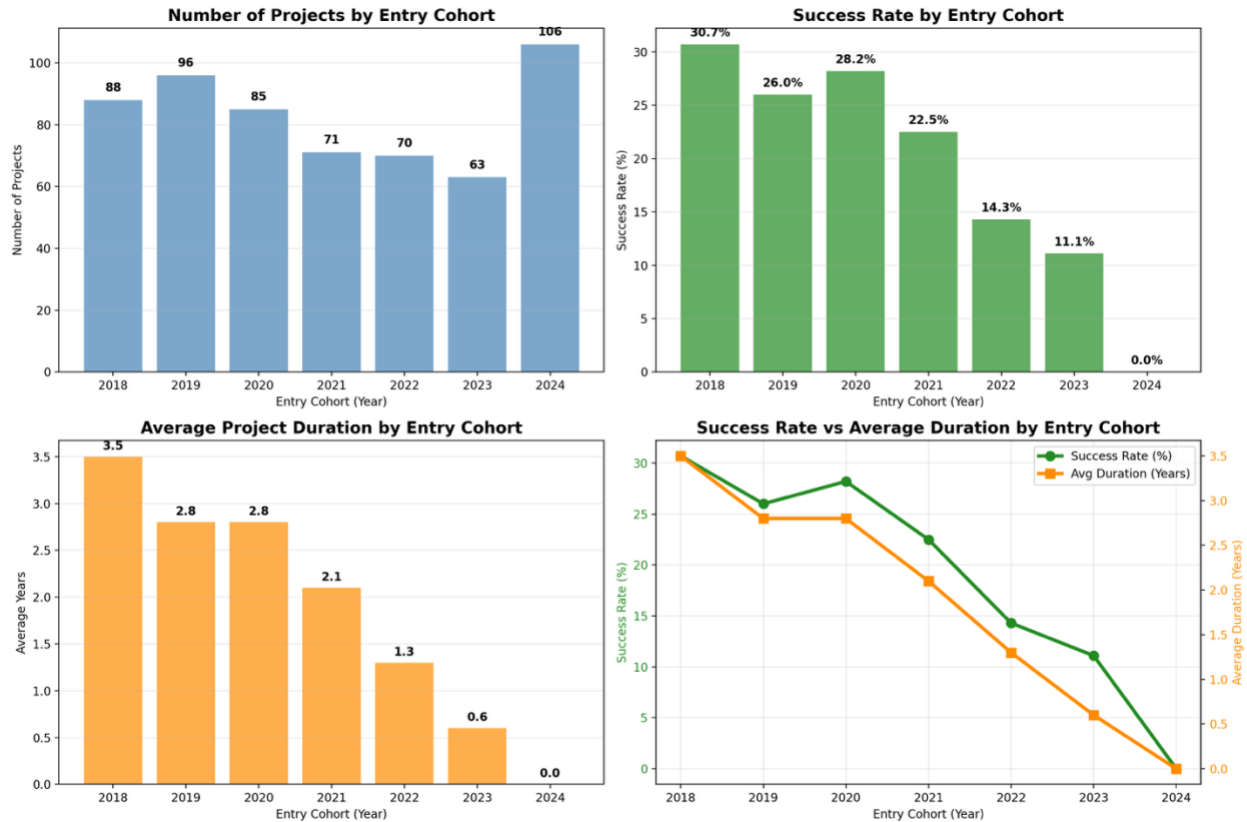


Figure 1: MPI Outcomes by Cohort.

Given the dataset's limited span, the 2018 cohort provides the most reliable base rate for project success, and progressions are modest. Figure 2 charts both overall and sector-specific outcomes for this cohort and plots project durations. Only 31 percent advance to construction within six years. The curves reveal steep early-stage attrition and underscore the persistent challenge of turning developmental milestones into shovels in the ground.

Six-Year Construction Outcomes: 2018 Cohort Analysis

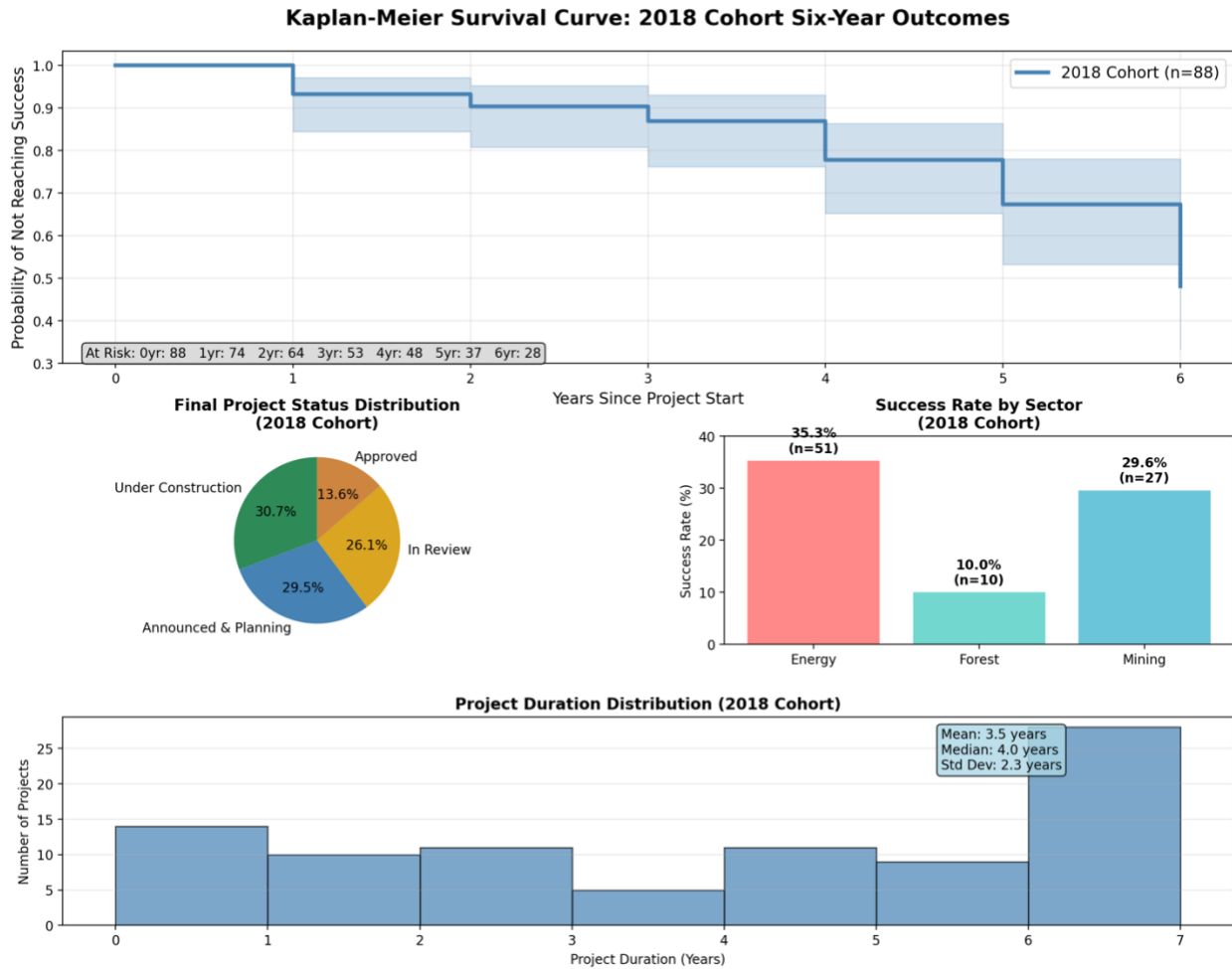


Figure 2: Six-Year Construction Outcomes of the 2018 Cohort.

Figure 3 documents substantial regional variation. Observed provincial variation in success rates appears closely tied to project mix, regulatory structure, and sample size, yet firm causal links remain unproven. Alberta's 32 percent completion record likely benefits from its concentration of midstream energy assets, a mature contractor base, and a well-established provincial review regime that aligns reasonably with federal requirements. Yukon, New Brunswick, and Prince Edward Island report even higher ratios, but these figures rest on very small cohorts dominated by lower-risk, locally sponsored projects. Ontario and Quebec introduce the greatest number of

proposals but progress fewer, a pattern that may reflect multi-layer municipal, provincial, and federal approvals that lengthen schedules and increase cost. British Columbia's 6 percent success rate may reflect the preponderance large, capital-intensive LNG and mining ventures facing rigorous environmental and Indigenous engagement processes.

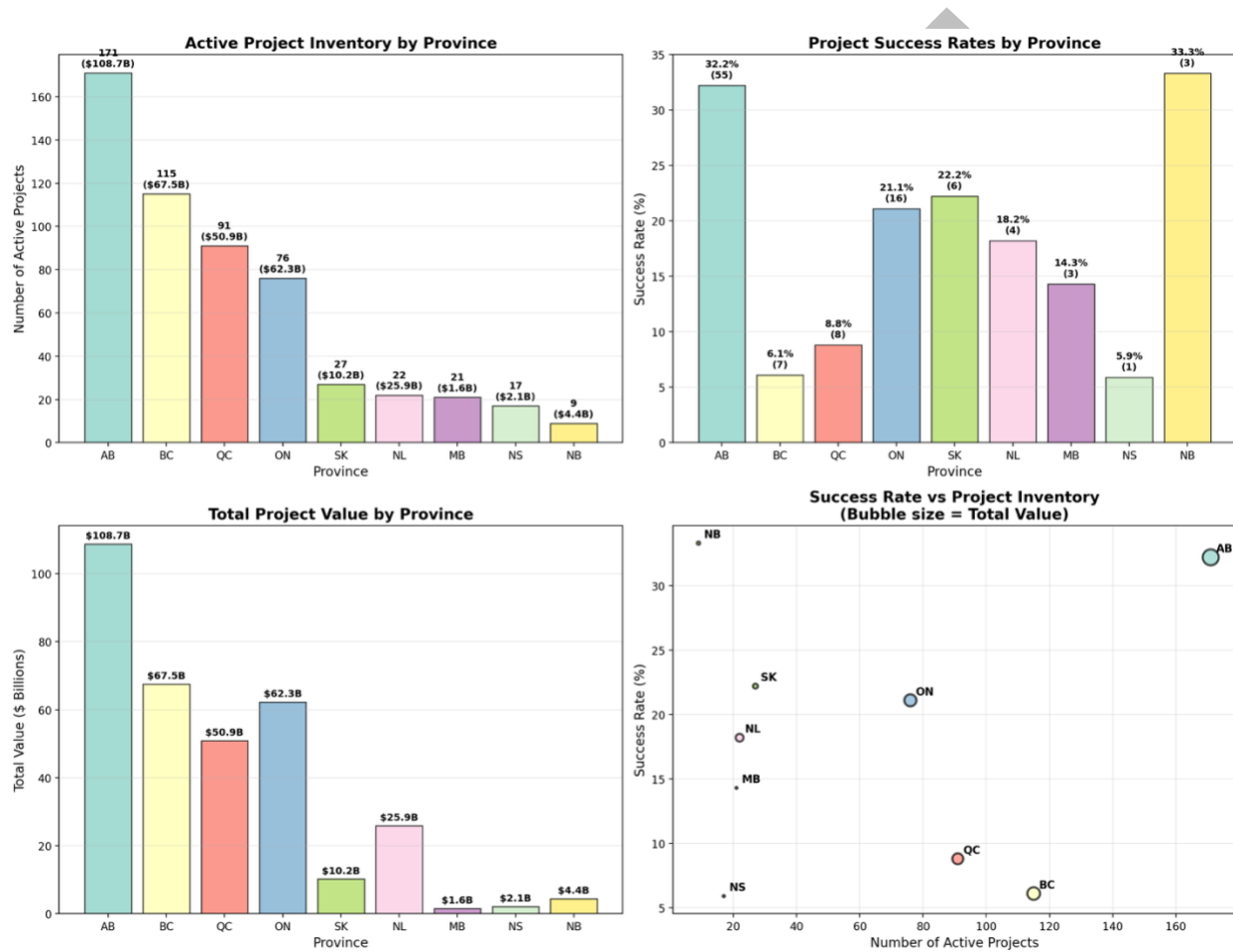


Figure 3: MPI Outcomes by Province.

Sectoral trajectories

Sectoral analysis in Figure 4 shows energy dominating both pipeline and outcomes: two-thirds of new starts and three-quarters of construction successes fall inside the energy category. Mining lags considerably. Only 10 percent of mining projects break ground versus 22 percent for energy

and 18 percent for forest-products facilities. Logistic regression indicates that, all else equal, a mining proposal is 60 percent less likely to succeed than an energy peer, underscoring sector-specific considerations such as commodity cycles, complex environmental reviews, and evolving Indigenous consultation protocols.

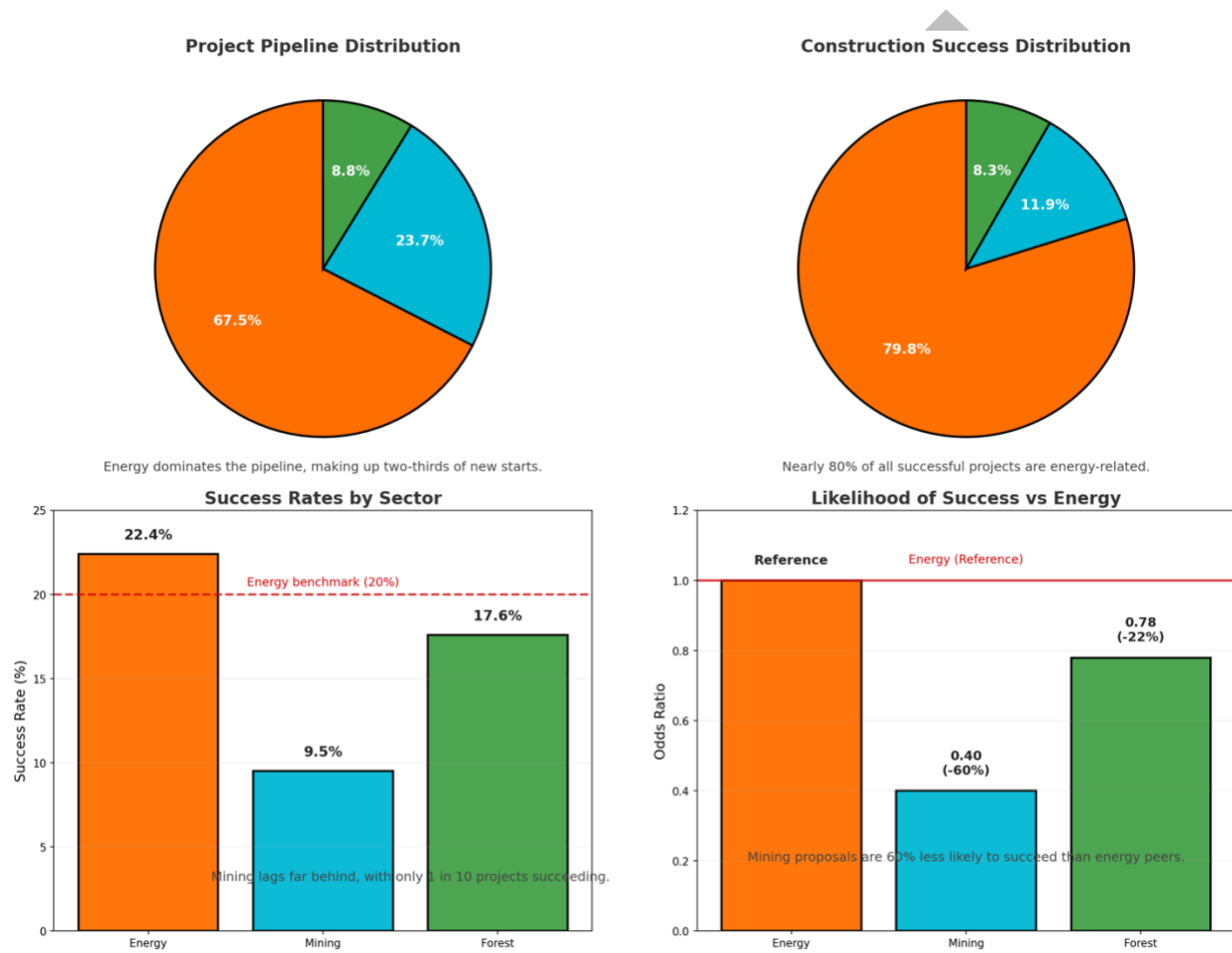


Figure 4: MPI Outcomes by Sector (Energy, Mining, Forestry).

Cost-tier outcomes

Figure 5 suggests that relationship between project scale and success is more complex than previously assumed. Conventional wisdom holds that the larger a project's capital budget, the lower its probability of advancing cleanly from announcement to operation. That said, the

relationship is not strictly linear. Small projects can also underperform because they lack economies of scale and professionalized governance. The MPI data indicate a more irregular distribution. Mega-projects exceeding \$5 billion continue to face substantial challenges, achieving a success rate of just 7 percent. In contrast, large projects valued between \$1.2 billion and \$5 billion perform best, with a 22 percent success rate, suggesting that this tier may offer a reasonable balance between ambition, financing capacity, and deliverability. Surprisingly, mid-sized projects (\$300 million to \$1.2 billion) underperform relatively, with a success rate of only 17 percent. Projects in the \$100 million to \$300 million range fare slightly better at 19 percent, while small projects under \$100 million achieve a 20 percent success rate. These zig-zag or sawtooth results challenge the assumption of a predictable relationship between scale and outcome and underscore the interplay of regulatory burden, capital availability, and execution risk across different investment brackets.

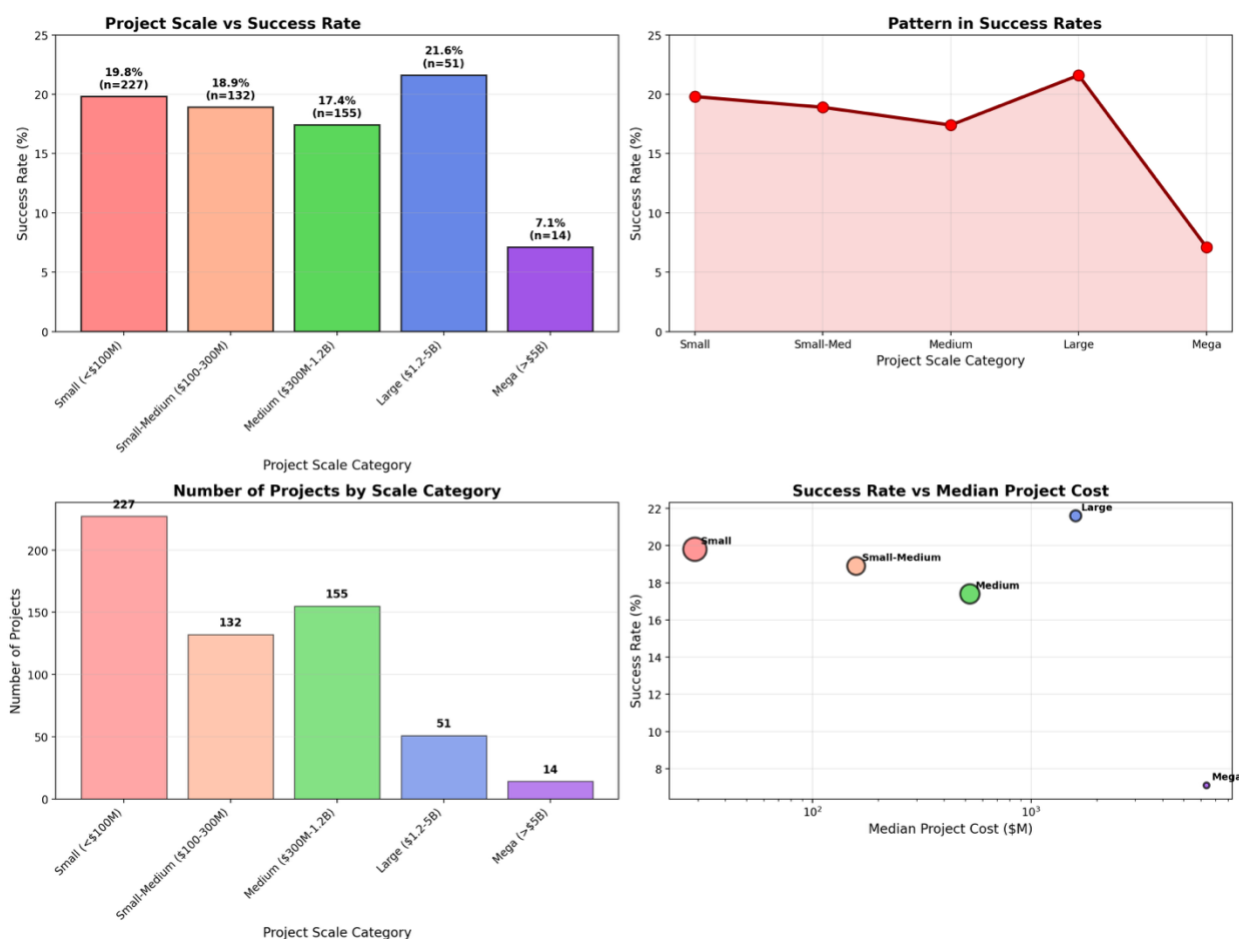
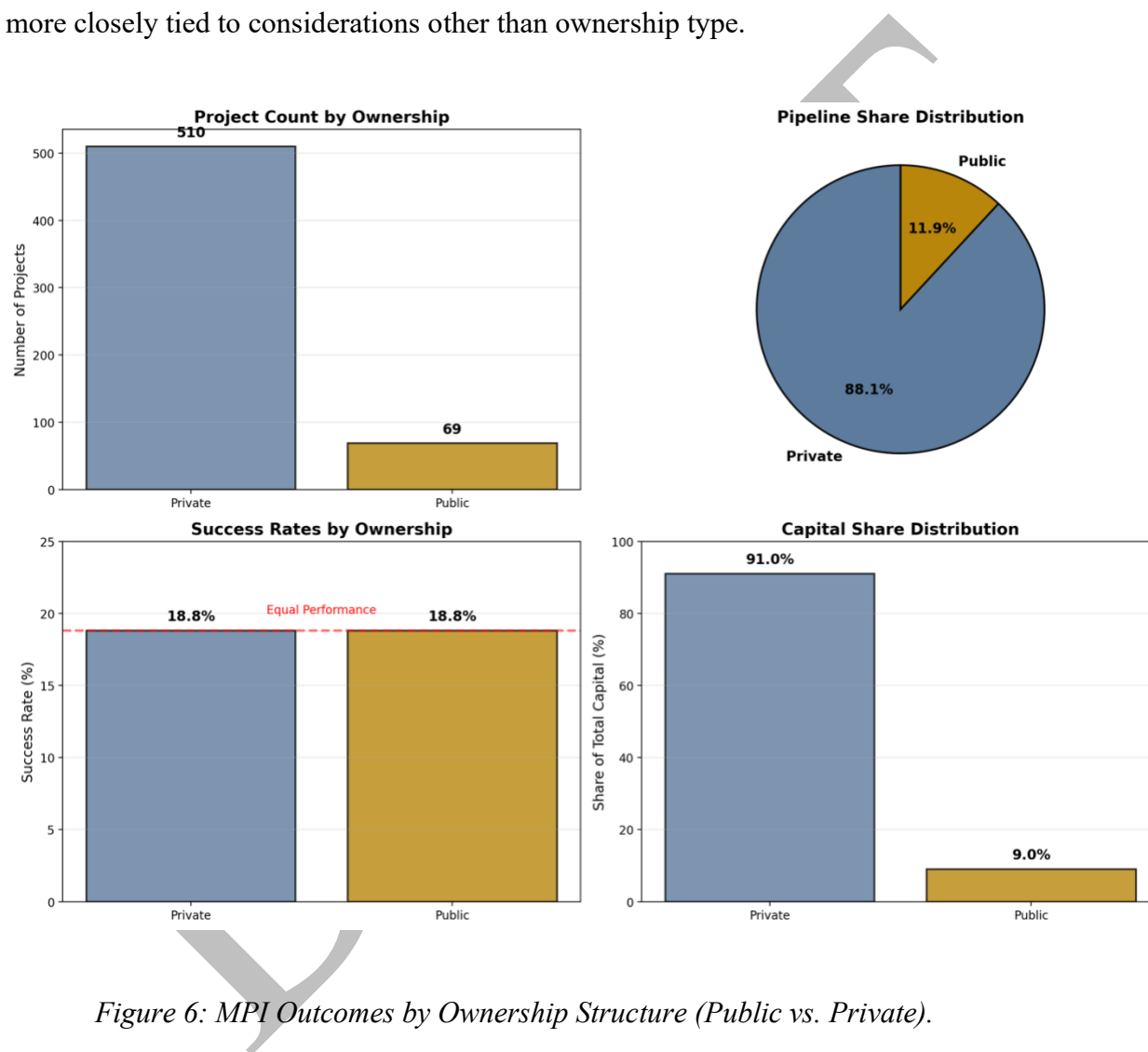


Figure 5: MPI Outcomes by Project Size.

Ownership structure

Ownership appears largely neutral in determining project outcomes, as indicated in Figure 6. Both private and public proponents post identical success rates of 19 percent (96 out of 510 private projects and 13 out of 69 public ones) indicating that ownership structure alone does not drive project success. However, private sponsors dominate the pipeline, accounting for 88 percent of all projects and 91 percent of total capital investment. Private projects also tend to be larger, with an average size of \$614.9 million versus \$448.5 million for public projects, and an even wider gap in median size (\$162.5 million vs \$110 million). Sectoral and regional variations

are more decisive: private firms outperform public ones in the energy sector (23 percent vs 19 percent), while both face challenges in mining. Regionally, private projects show high success in Alberta (32 percent) but struggle in British Columbia (5 percent), where public ventures post a comparatively strong 20 percent success rate. These findings suggest that project performance is more closely tied to considerations other than ownership type.



Development-stage flows

Figure 7 follows 579 projects through the development pipeline and illustrates the flow and fate of projects. Of the 404 projects that begin in Announced & Planning, 265 (66 percent) remain there, while 85 (21 percent) move directly to construction, 28 shift to Approved, and 26 enter In

Review. The In Review stage is similarly sticky: 93 of 129 projects (72 percent) stay in review, with only 13 securing moving to Approved and 14 advancing directly to Under Construction. The Approved stage offers no guarantee of progress, either, as 36 of 46 projects (78 percent) start and end there and just 10 projects advance to Under Construction. Overall, 109 (19 percent) projects of the total reach Under Construction. The leapfrogging of intermediate stages may reflect deliberate streamlining, strategic readiness or data limitations.

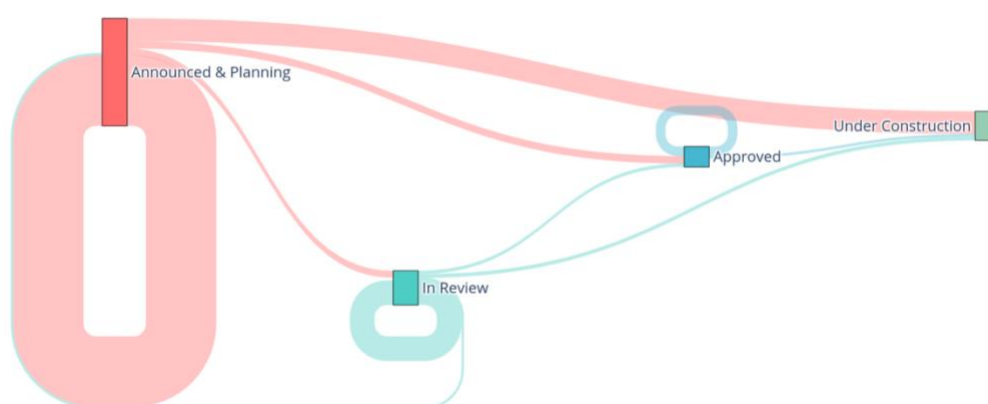


Figure 7: MPI Project Development Flow, 2018 to 2024.

Approval bottleneck

Among the 77 projects whose final MPI status is Approved, 29 (38 percent) have lingered in that category for three years or more, immobilizing roughly C\$14.3 billion in prospective capital. The pattern in Figure 8 is most pronounced in energy (16 projects) and mining (13 projects) and spans several jurisdictions, notably Nova Scotia, Ontario, Alberta, and Newfoundland and Labrador. On average, these projects remain in limbo for 4.2 years, underscoring that regulatory clearance, while necessary, is not sufficient to trigger construction. Market volatility, financing hurdles, and policy uncertainty can all erode project economics after approvals are secured,

leaving sponsors unable or unwilling to proceed. This persistent post-approval bottleneck carries significant economic costs. Understanding its root causes could unlock billions in investment and accelerate project delivery.



Figure 8: MPI Projects Stuck Approval for 3+ Years.

Policy era effects

Over the study period, the Canadian federal government introduced four major policy initiatives to mobilize development of major projects. First, the implementation of the Canada Infrastructure Bank (CIB) in 2018 established a \$35 billion vehicle offering below-market debt

and credit enhancements to companies, provinces, territories, municipalities and Indigenous communities that sponsor revenue-generating projects. After a two-year start-up phase, the CIB's deal flow increased noticeably between 2020 and 2023. Second, Budget 2021 launched the \$1.5 billion Clean Fuels Fund (CFF) primarily to support the construction of clean fuel production facilities, although initial contribution awards were not announced until 2022. Third, the 2022-23 fiscal cycle unveiled refundable Clean Energy Investment Tax Credits (CE-ITCs) covering 15 percent to 40 percent of eligible costs for renewable equipment, clean electricity, and hydrogen. Finally, the \$15-billion Canada Growth Fund (CGF) became operational in 2023, using equity and carbon contracts for difference to crowd in private capital, expand Canada's clean-fuel supply, and help decarbonize hard-to-abate sectors such as oil and gas and heavy industry. Collectively, these instruments are designed to accelerate projects from early concept to construction.

The four-panel dashboard in Figure 9 offers a provisional view of policy traction along the project pipeline. Panel 1 shows announcements rebounding in 2024 after a six-year slide, while Panel 2 records more than \$100 billion in proposed capital the same year, spikes that coincide with the CGF's launch and the publication of Clean-Energy ITC rules. By contrast, Panel 3 underscores a stubborn bottleneck: early-stage progression has continued to fall, posting zero for the most recent cohort. Panel 4 suggests approvals are moving faster, with median timelines now half those seen during the CIB start-up era. Taken together, the data imply that federal initiatives seem to be helpful in mobilizing capital and shortening reviews, yet more time and study is required to understand whether early-stage momentum converts into higher progression rates. Certainly the 2024 surge in announcements and project size hints that the policy suite may be

gaining traction, but confirmation will require additional cohorts to work through the development pipeline.

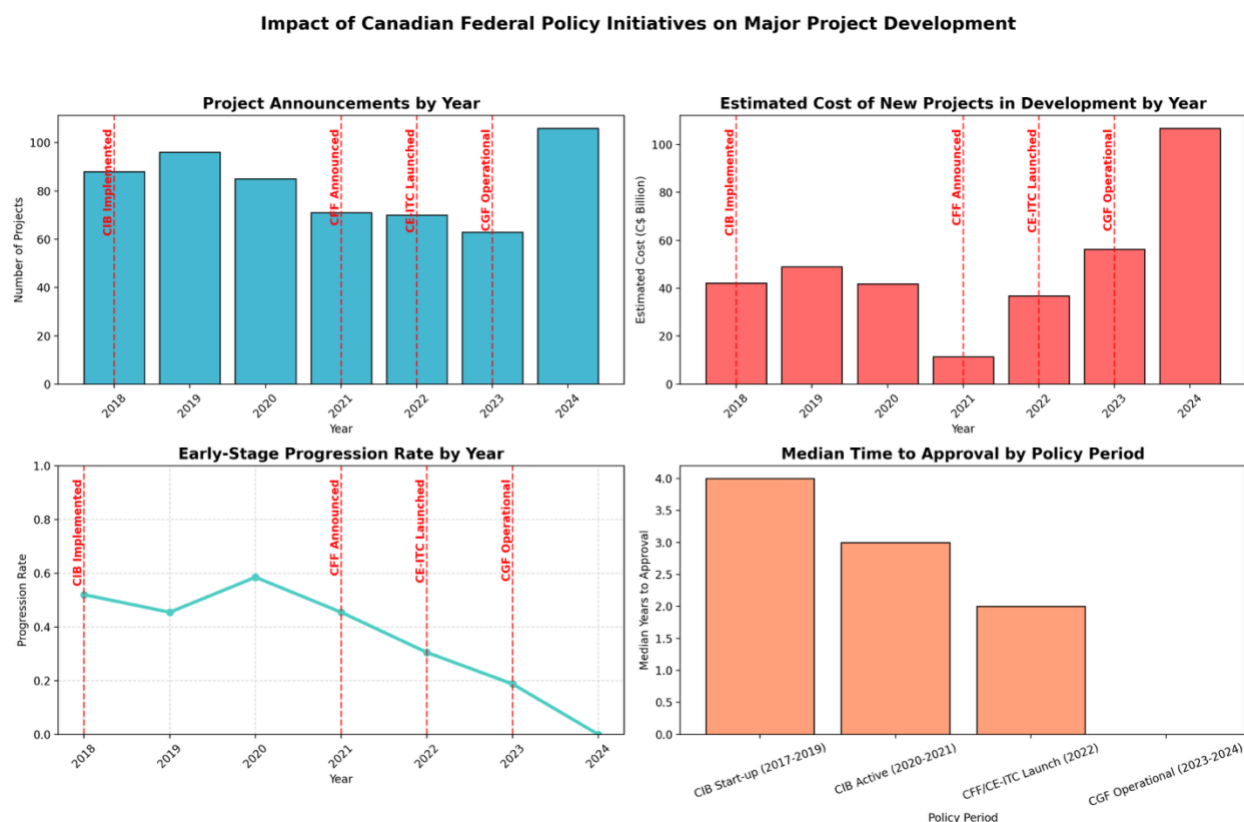


Figure 9: Impact of Major Canadian Policy Initiatives on MPI Projects.

Conclusion

This paper provides a cross-sectoral, cross-provincial and longitudinal assessment of Canada's major project pipeline using data from the Major Projects Inventory (2018–2024). Despite over half a trillion dollars in planned capital investment, only 19 percent of tracked projects advanced to construction. Most projects stall early: two-thirds of those in the Announced and Planning stage never progress. But bottlenecks persist even after regulatory clearance, with 38 percent of Approved projects appearing stuck and inactive for three years or more.

Outcomes vary by sector and region. Energy projects perform better than mining or forest-sector projects, and Alberta outpaces other provinces in completion rates. Large scale projects tend to be more successful than mid-size projects, while both mega and small initiatives face added hurdles. Although ownership type appears neutral, private firms dominate the pipeline by volume and value. Federal policy and incentive initiatives appear to coincide with a rise in project announcements and project size. Ongoing monitoring of newer cohorts, however, is critical to see whether these measures translate into higher progression rates or if additional, more targeted interventions will be required to overcome lingering bottlenecks and unlock the full potential of Canada's infrastructure pipeline.